Kinvecta

A KINVECTA SERIES

How Large Language Models Work

From math to transformers

EPISODE 09 · GRADIENT DESCENT, PART 1



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HOW LARGE LANGUAGE MODELS WORK · EPISODE 9 ·

Three hikers. One slope.

Same hill, same direction. Only the size of the stride is different, and it decides everything.

17-minute lesson

Companion notebook



MARK

THE STRIDE

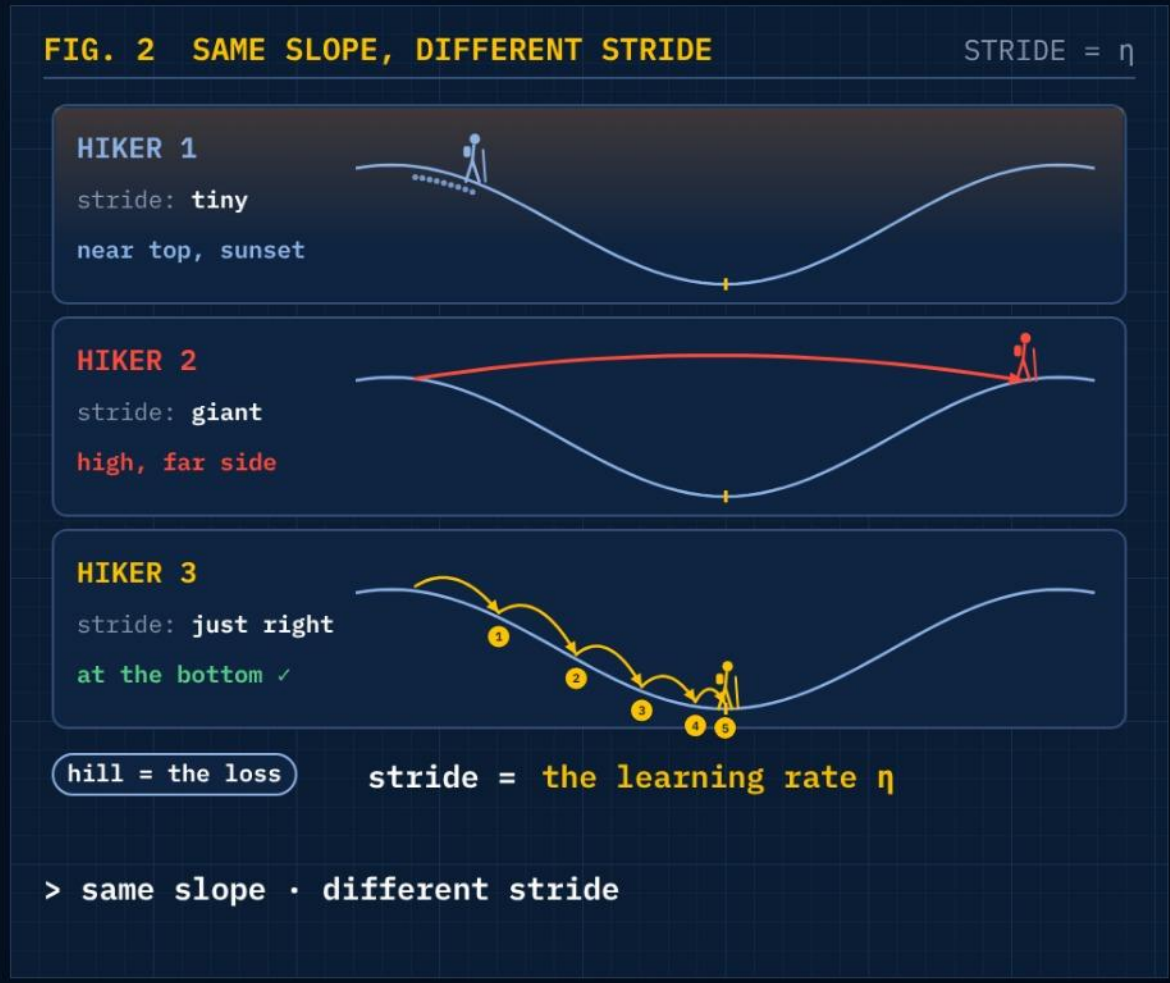
The stride decides everything

A model learning walks downhill on its loss, like the hikers.

01 Same slope, same direction

02 Only the size of the stride differs

03 That stride is the learning rate



WHY THIS MATTERS

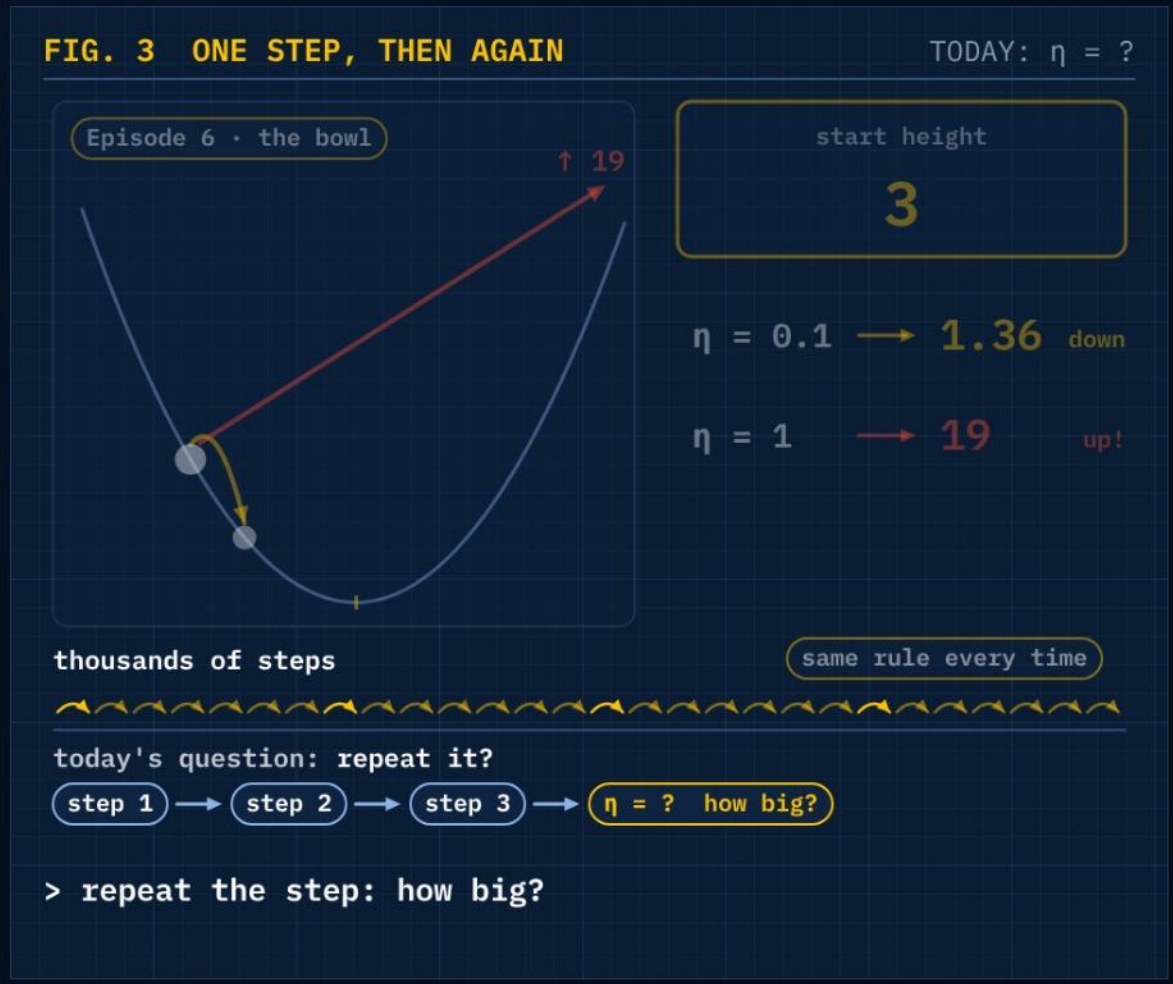
One step, then thousands

Episode 6 took one step down a bowl, from height 3.

01 $\eta = 0.1 \rightarrow 1.36 \cdot \eta = 1 \rightarrow 19$

02 Training repeats the same step thousands of times

03 Repeat it: how big should it be?



Episode 6 (the gradient and the step). Worked example; every number is checked in the companion notebook.

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EPISODE 9 · GRADIENT DESCENT, PART 1

The learning rate

Three parts, every step worked by hand, and a notebook with all of them.

01 The update rule, one piece at a time

02 One knob: every step by hand

03 The bowl: who sets the speed limit

04 Plus a myth about playing it safe



Companion notebook (Colab), linked in the description. Sources on every slide.

MARK

A QUICK REMINDER

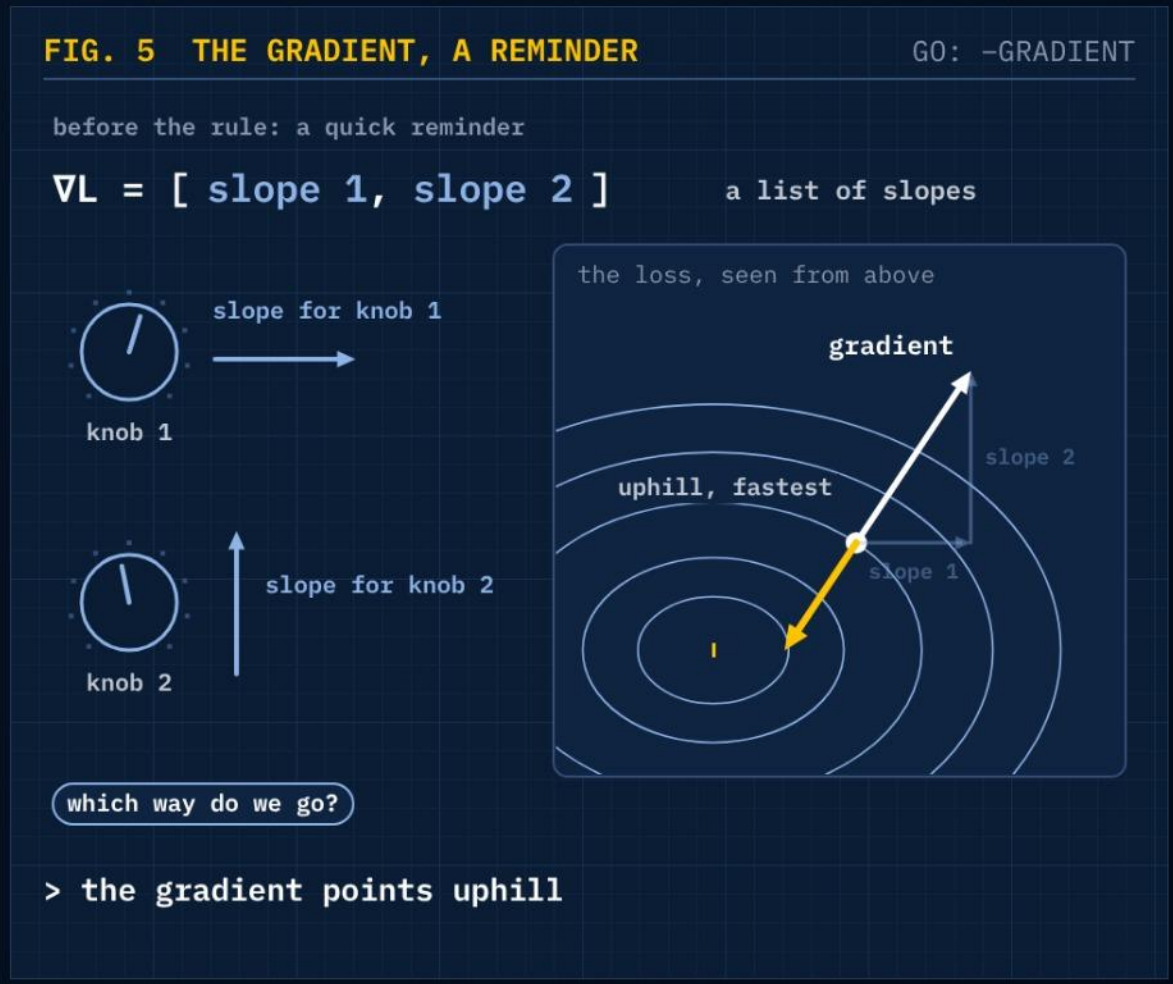
What a gradient is

One slope for every knob.

01 Each slope: nudge this knob, how fast does the loss rise?

02 Together they point uphill, fastest

03 So we walk the opposite way



Episode 6 (the gradient). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization, eq. 4.5).

PART 1 · THE RULE

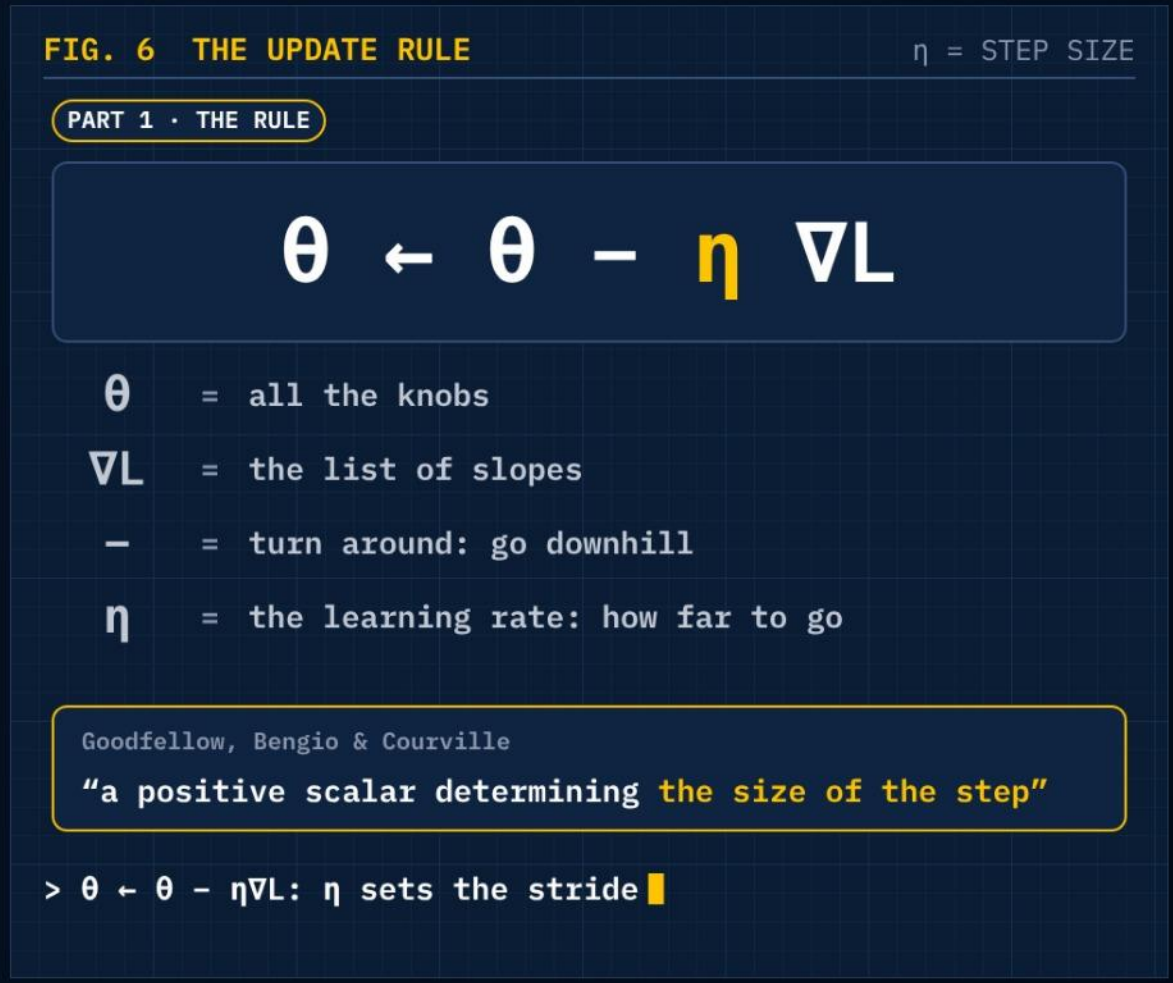
$\theta \leftarrow \theta - \eta \nabla L(\theta)$

Read it one piece at a time.

01 θ: all the knobs · ∇L: the list of slopes

02 The minus sign turns us downhill

03 η, the learning rate: “a positive scalar determining the size of the step”



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization, eq. 4.5).

PART 1 · THE RULE

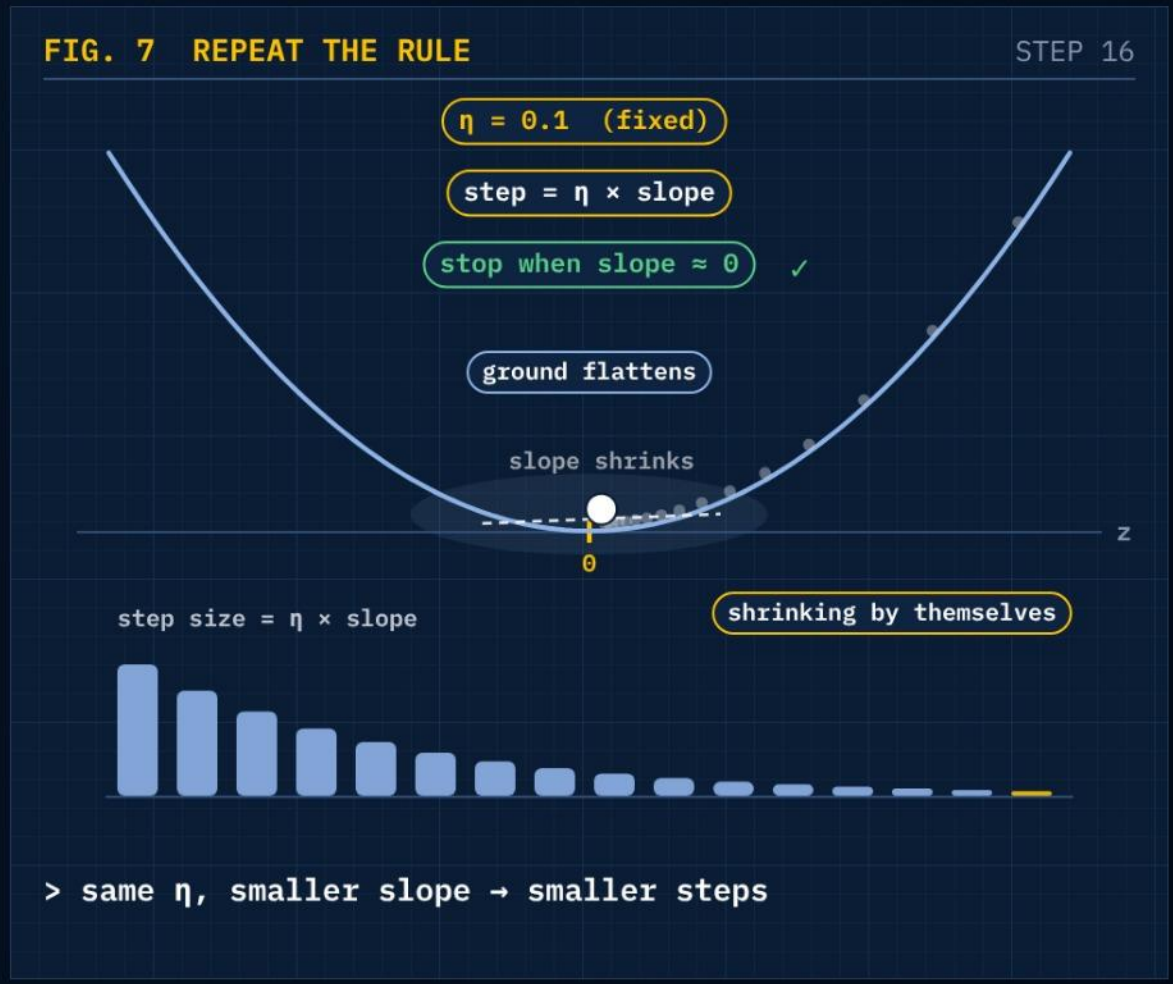
Again, and again

Stop when the gradient is zero, or very close.

01 Repeat until the slope is about zero

02 Step = η × slope

03 Near the bottom, steps shrink by themselves



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization, eq. 4.5).

PART 1 · WHY H NEEDS CARE

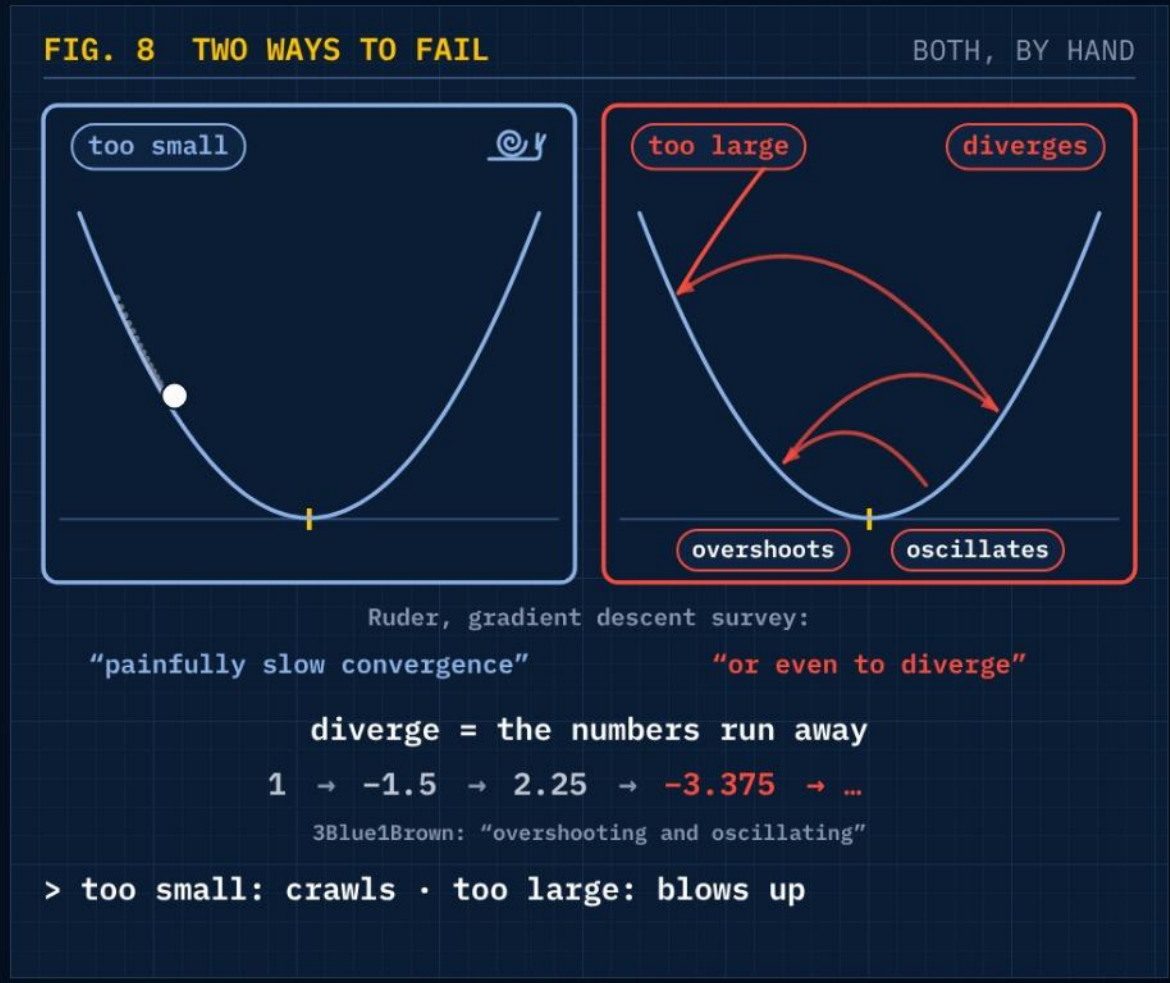
Two ways to fail

Too small, or too large.

01 Too small: “painfully slow convergence”

02 Too large: fluctuate, “or even to diverge”

03 Diverge: the numbers run away



Ruder, “An overview of gradient descent optimization algorithms” (arXiv:1609.04747, 2016), §§2–3. 3Blue1Brown, “Gradient descent, how neural networks learn,” *Neural networks*, ch. 2 (2017).

PART 2 · ONE KNOB

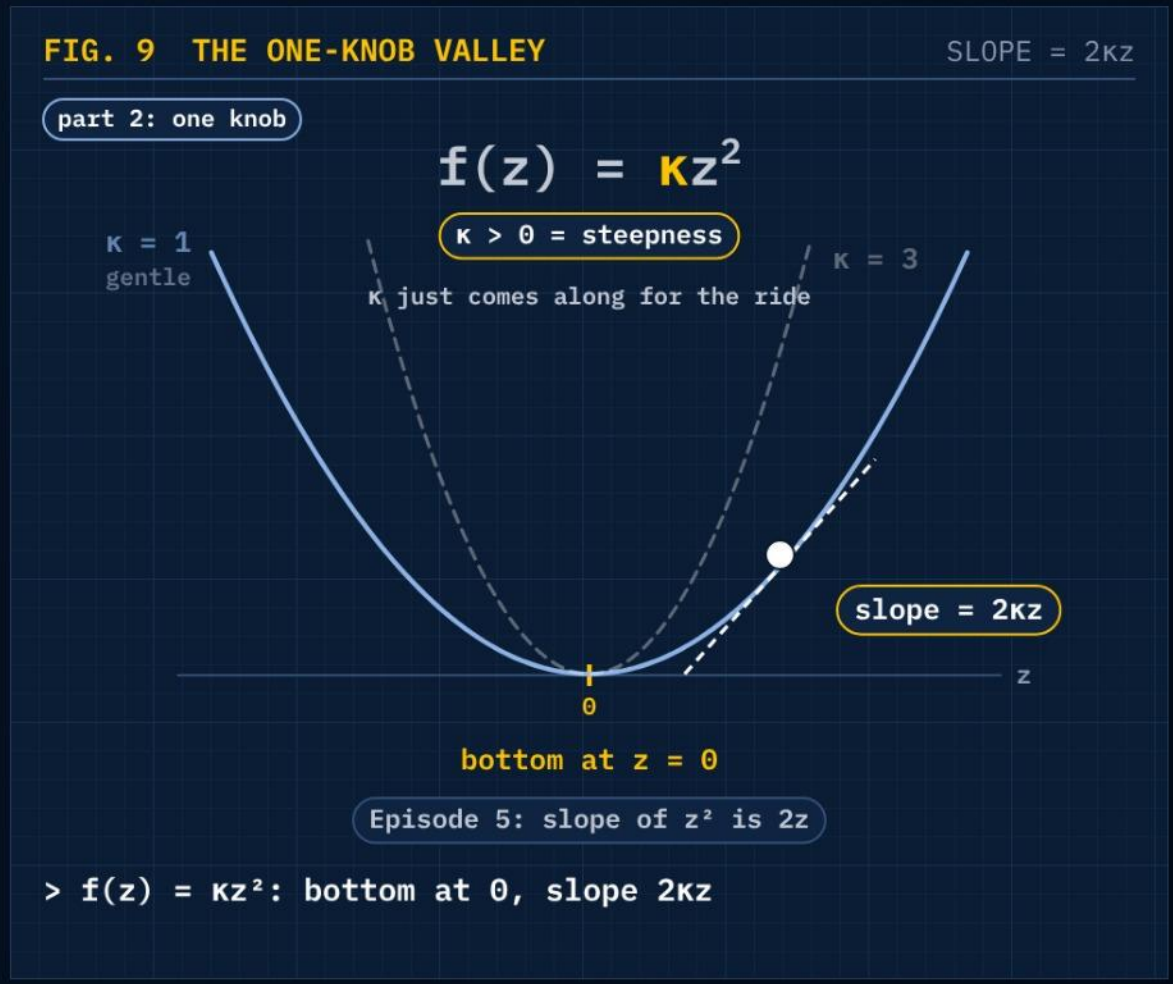
The simplest valley

$f(z) = \kappa z^2$

01 z is the knob; the bottom is at z = 0

02 κ > 0: how steep the walls are

03 The slope is 2κz (Episode 5: z^2 has slope 2z)



PART 2 · ONE STEP, SLOWLY

The factor

$z_{new} = z - \eta \cdot 2\kappa z$

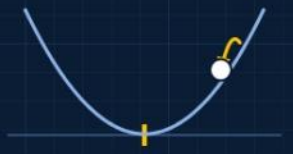
01 New z = old z – η × slope

02 Pull out the z

03 z_new = (1 – 2ηκ) · z

FIG. 10 ONE STEP, SLOWLY — THE FACTOR

one step, slowly



$$\text{new } z = \text{old } z - \eta \cdot \text{slope}$$

$$\text{slope} = 2\kappa z$$

$$\text{new } z = z - \eta \cdot 2\kappa z$$

$$\text{new } z = \boxed{(1 - 2\eta\kappa)} \cdot z$$

the factor

the most important number in this episode

> one step multiplies z by (1 – 2ηκ)

Worked example; every number is checked in the companion notebook.

PART 2 · WHAT THE FACTOR MEANS

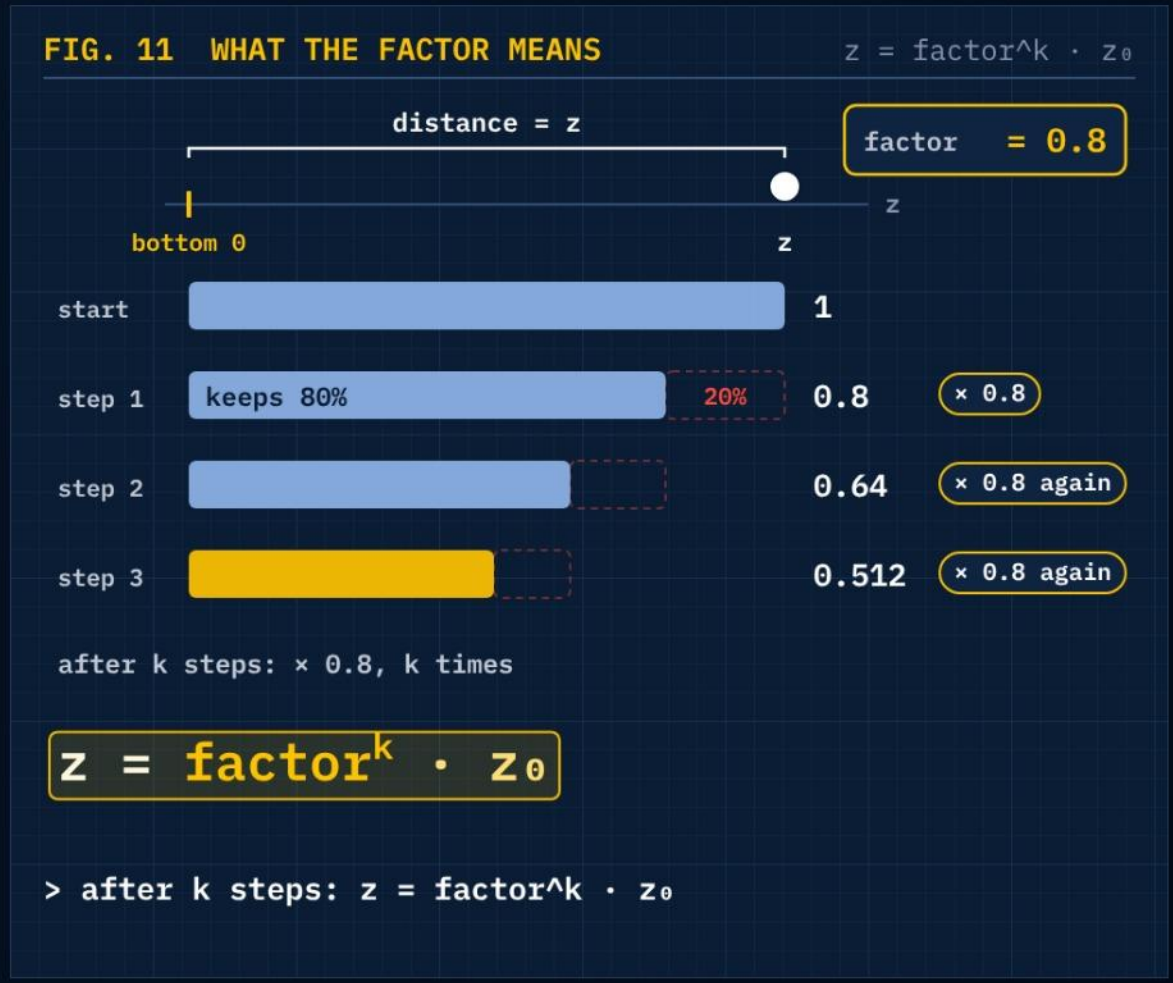
Keep a fraction, every step

z is your distance from the bottom.

01 z is the distance to the bottom

02 Factor 0.8: keep 80%, remove 20%

03 After k steps: $z = \text{factor}^k \cdot z_0$



Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND · K = 1, Z_0 = 1

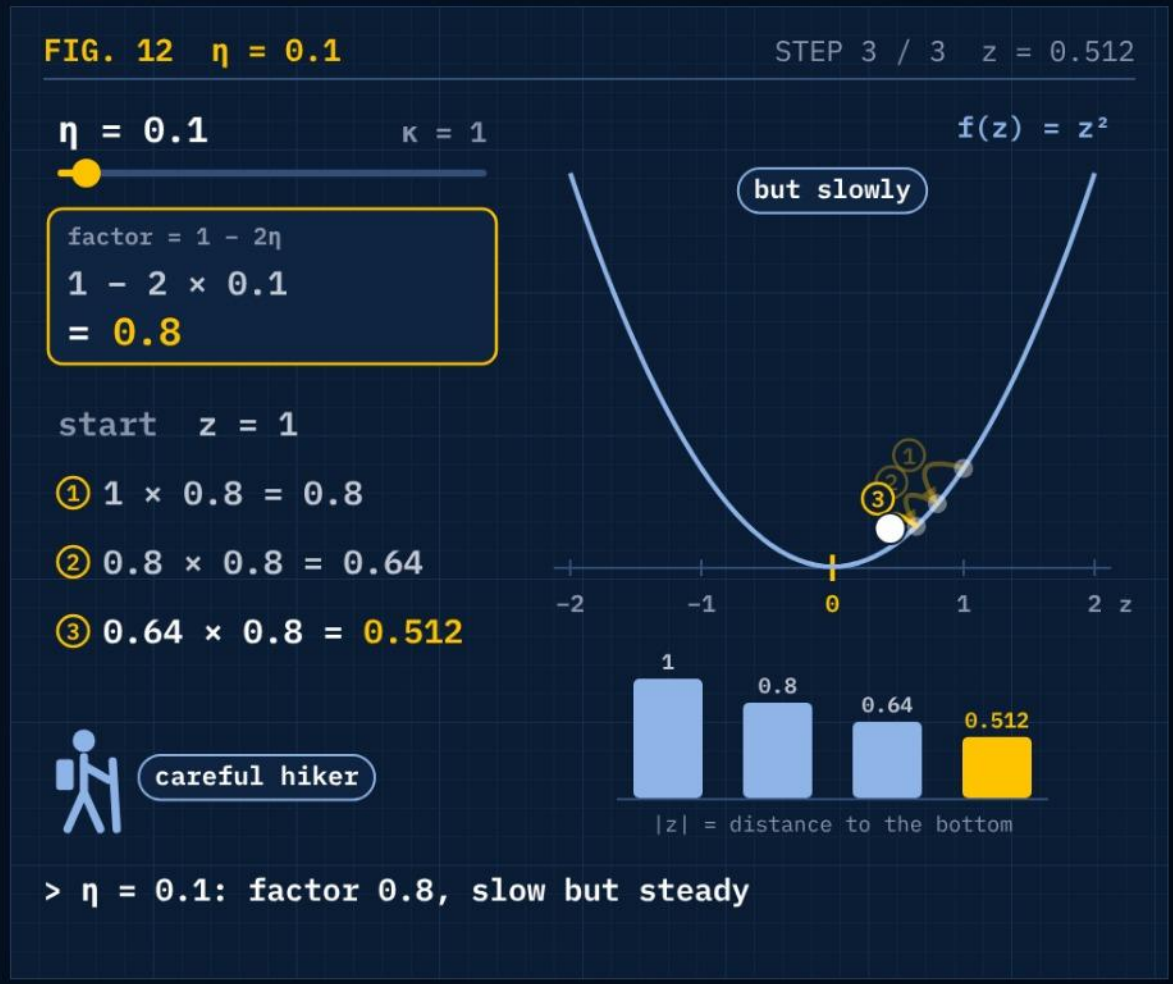
η = 0.1: slowly

Factor = 1 − 2 × 0.1 = 0.8

01 Factor 0.8

02 0.8 → 0.64 → 0.512

03 Closer every step, but slowly



Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND · K = 1, Z_0 = 1

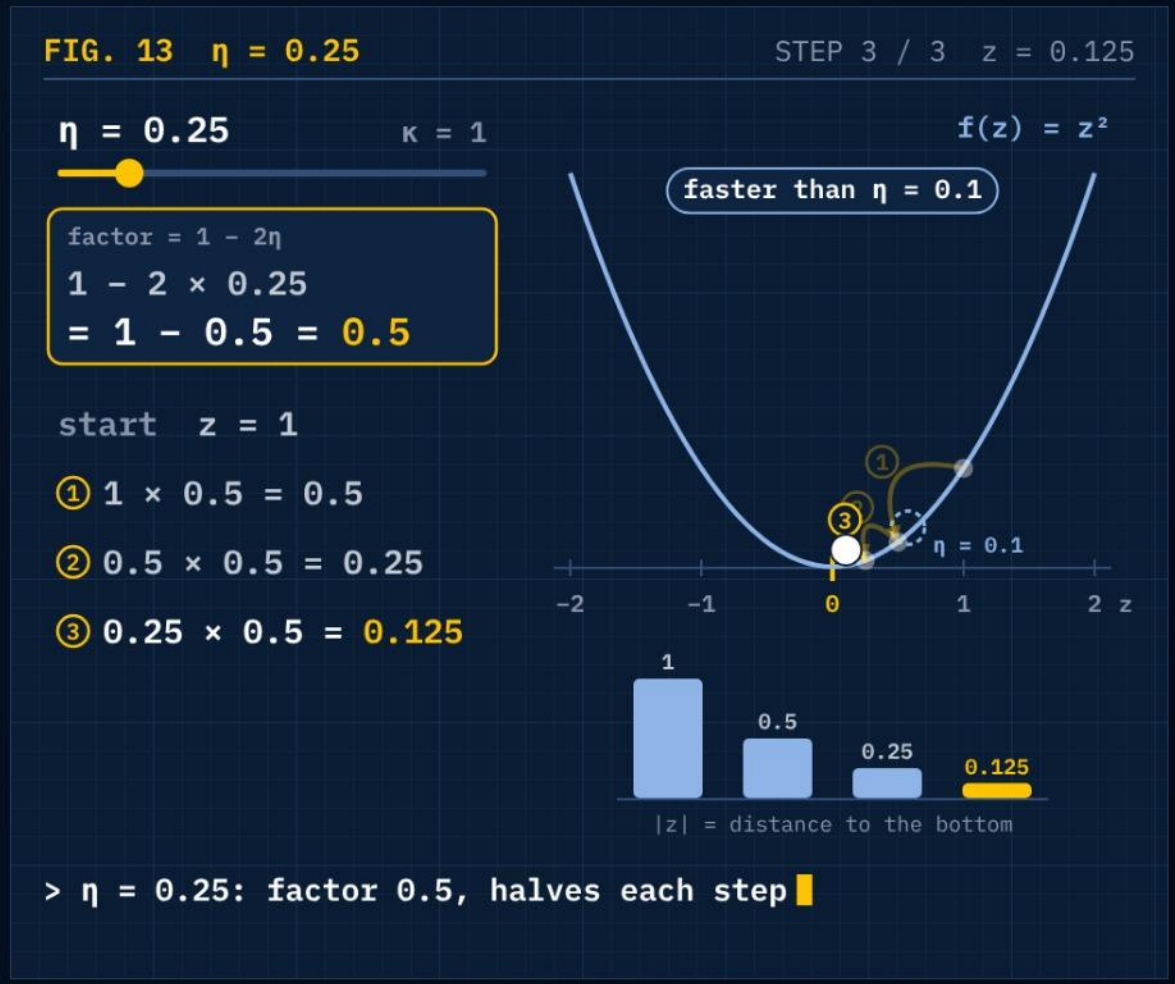
η = 0.25: halving

Factor = 1 − 0.5 = 0.5

01 Factor 0.5

02 0.5 → 0.25 → 0.125

03 Half the distance, every step



Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND · K = 1, Z_0 = 1

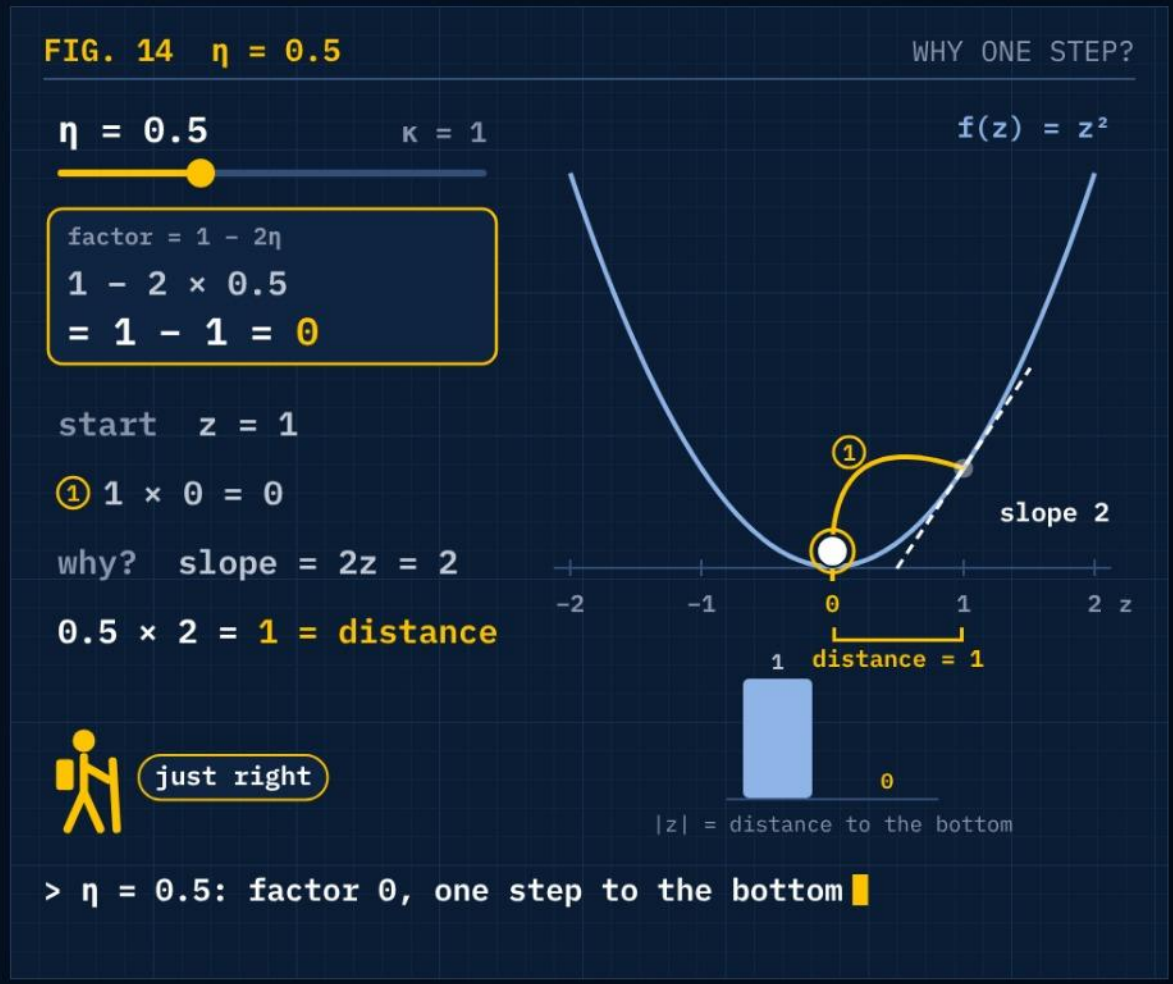
η = 0.5: one step

Factor = 1 − 1 = 0

01 Factor 0

02 One step, exactly at the bottom

03 Slope 2 × η 0.5 = 1 = the distance



Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND · K = 1, Z_0 = 1

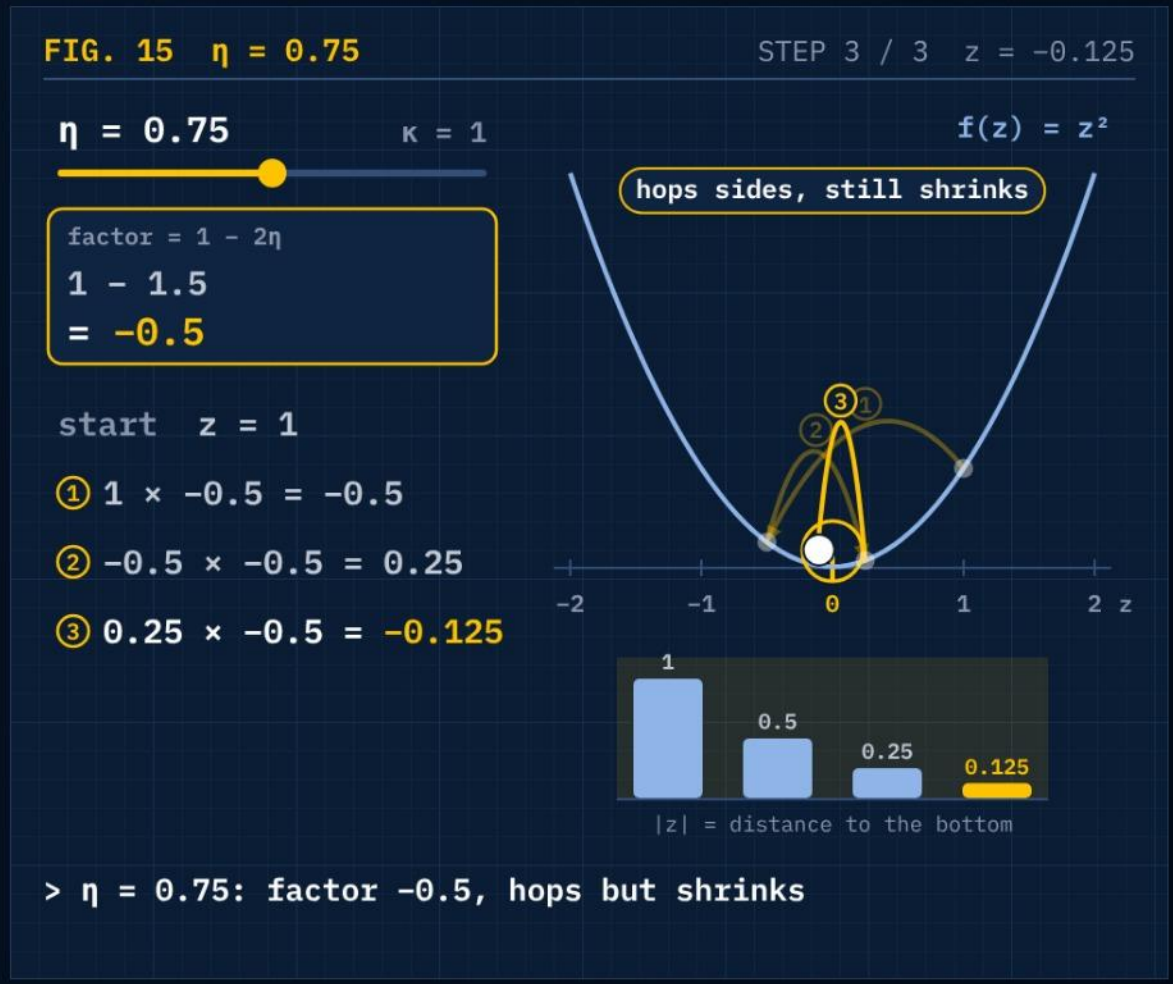
η = 0.75: hop and shrink

Factor = 1 − 1.5 = −0.5

01 Factor −0.5

02 Negative: land on the other side

03 −0.5 → 0.25 → −0.125: still arrives



Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND · K = 1, Z_0 = 1

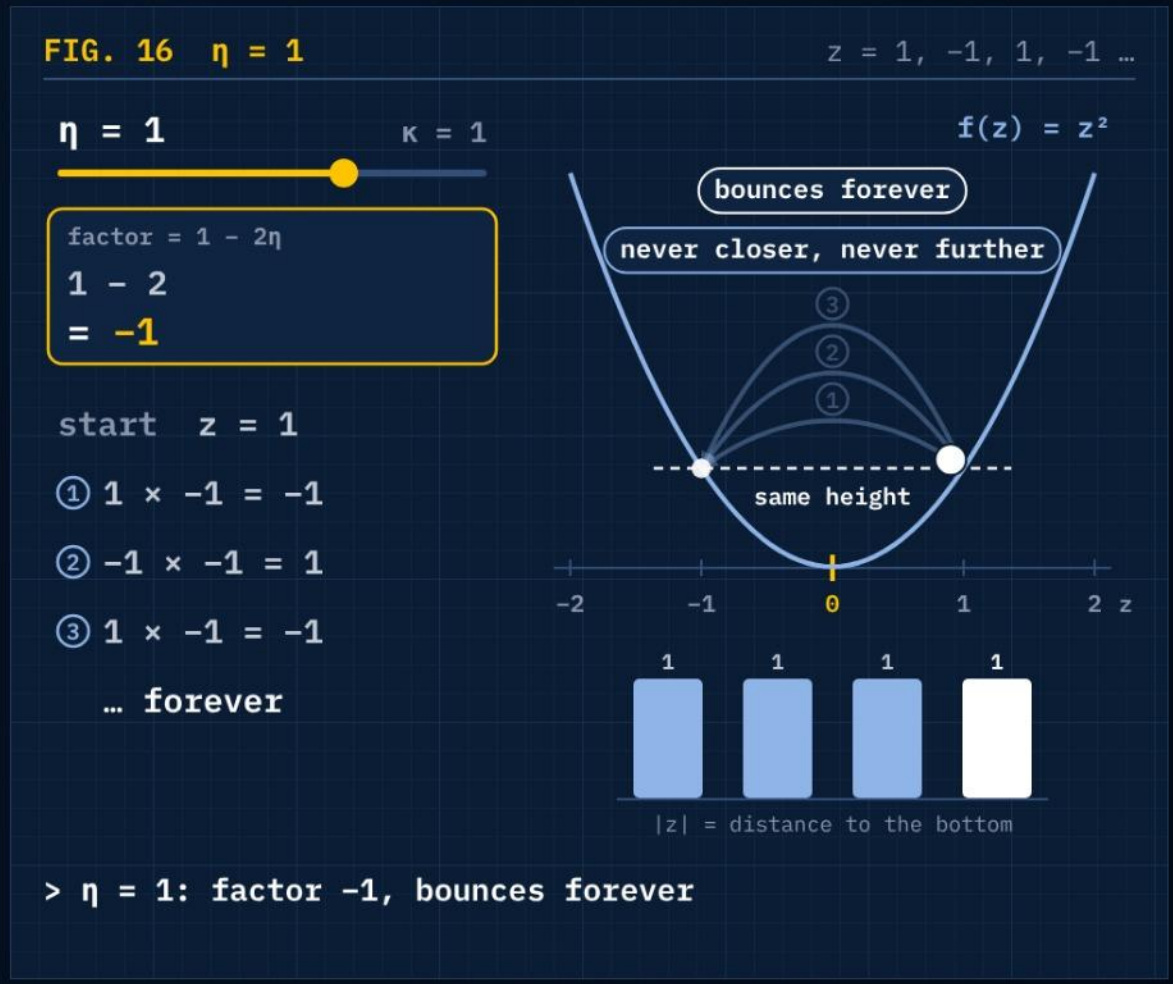
η = 1: bounce forever

Factor = 1 − 2 = −1

01 Factor −1

02 1 → −1 → 1 → −1

03 Never closer, never further



Worked example; every number is checked in the companion notebook.

PART 2 · BY HAND · K = 1, Z_0 = 1

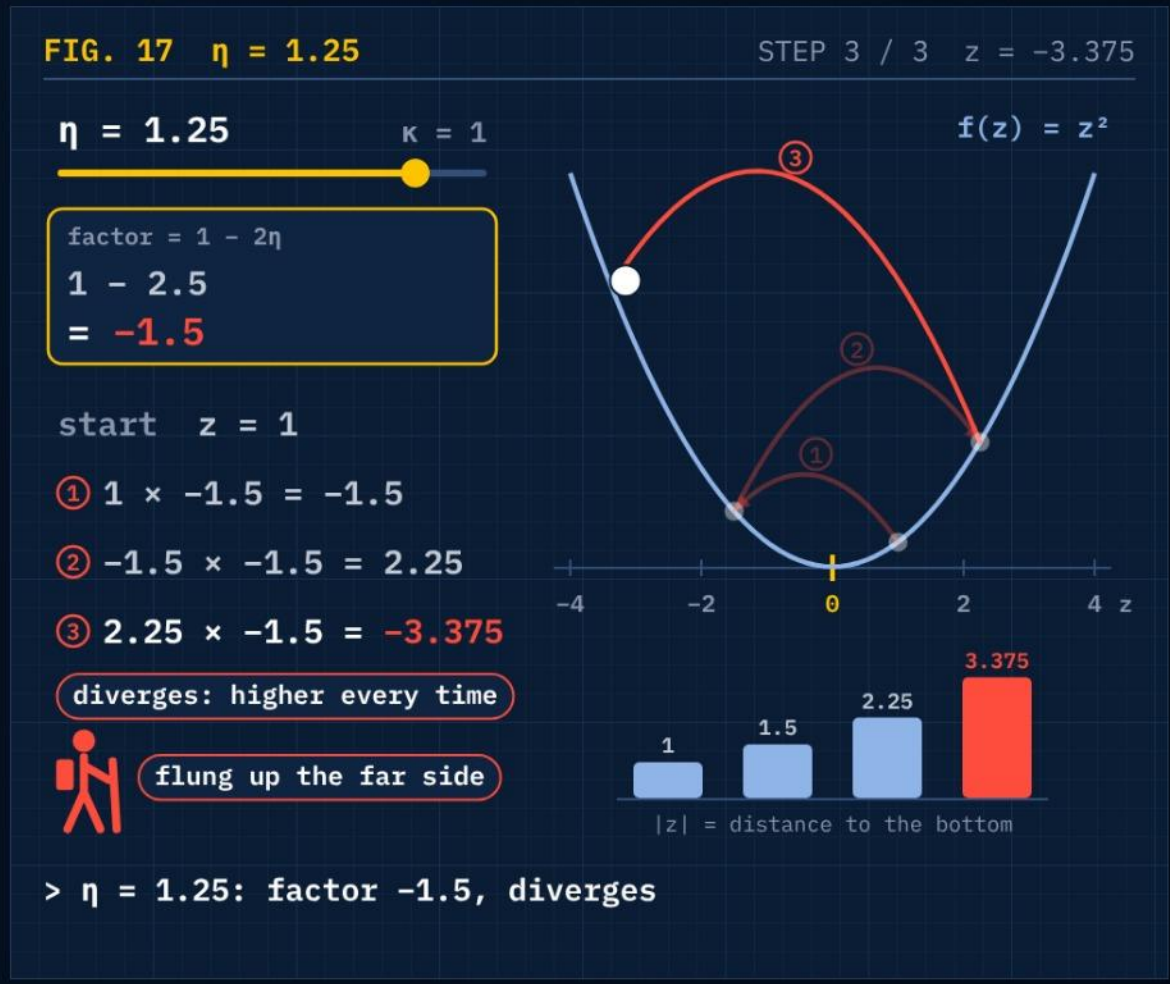
η = 1.25: blow up

Factor = 1 − 2.5 = −1.5

01 Factor −1.5

02 −1.5 → 2.25 → −3.375

03 Further every step: divergence



Worked example; every number is checked in the companion notebook.

PART 2 · THE WHOLE STORY

Settles when 0 < η < 1/κ

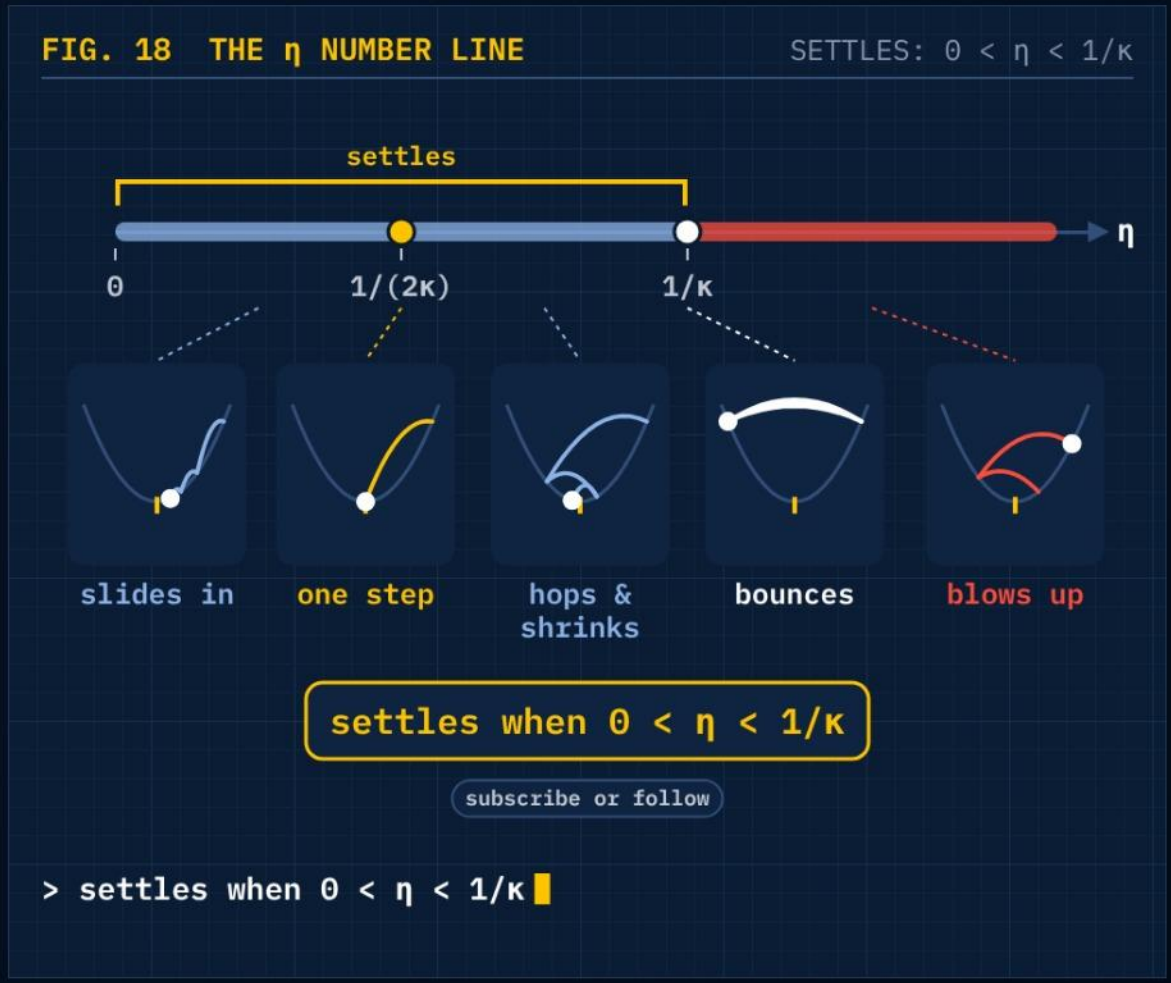
One number line for every step size.

01 Below 1/(2κ): slide in · at 1/(2κ): one step

02 Up to 1/κ: hop and shrink

03 At 1/κ: bounce · above: blow up

04 Settles exactly when 0 < η < 1/κ



Worked example; every number is checked in the companion notebook.

PART 2 · HOW SLOW IS SAFE?

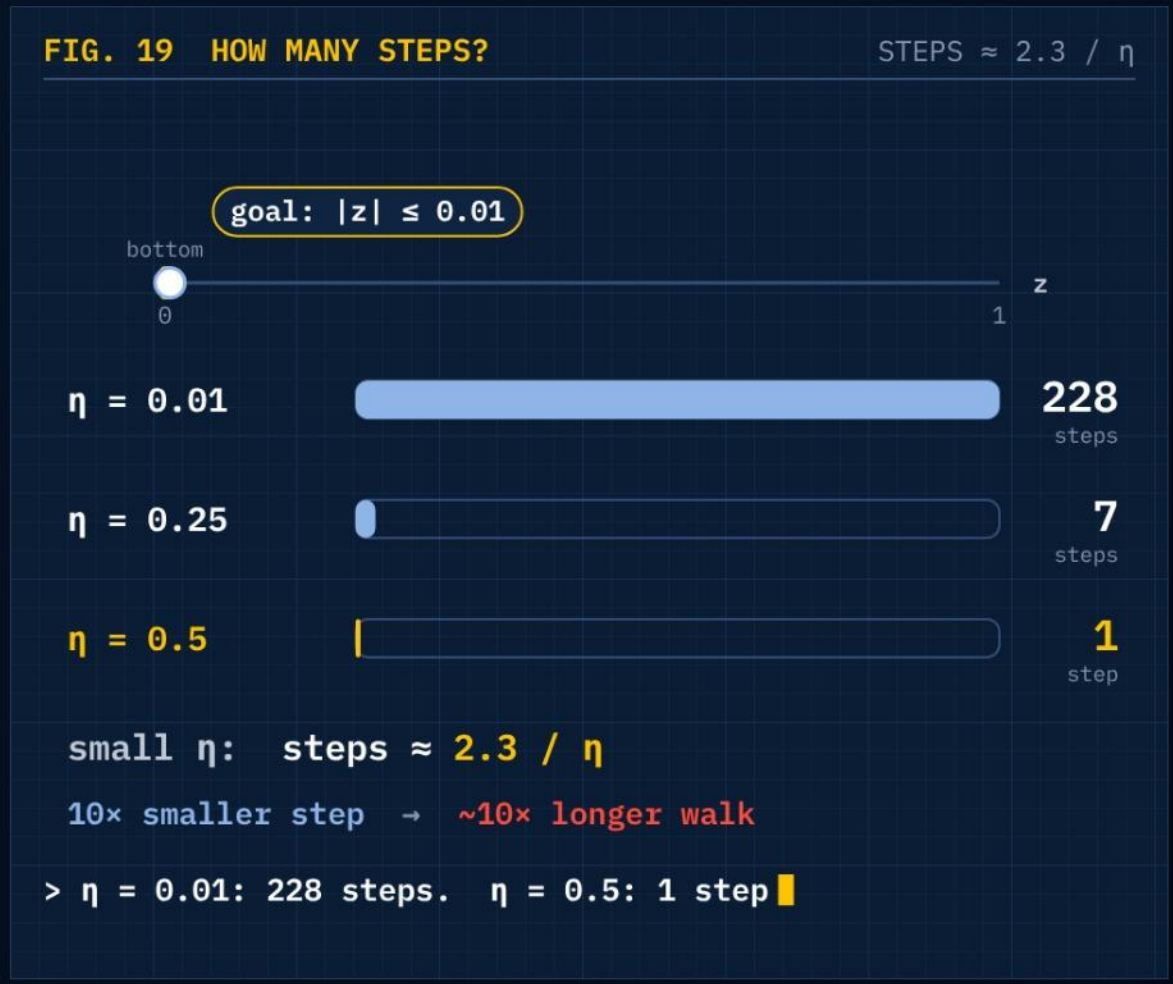
Counting steps

Steps to get within 0.01 of the bottom, from 1.

01 η = 0.01: 228 · η = 0.25: 7 · η = 0.5: 1

02 Small η: about 2.3 / η steps

03 10× smaller step, about 10× longer walk



Worked example; every number is checked in the companion notebook.

PART 3 · THE BOWL

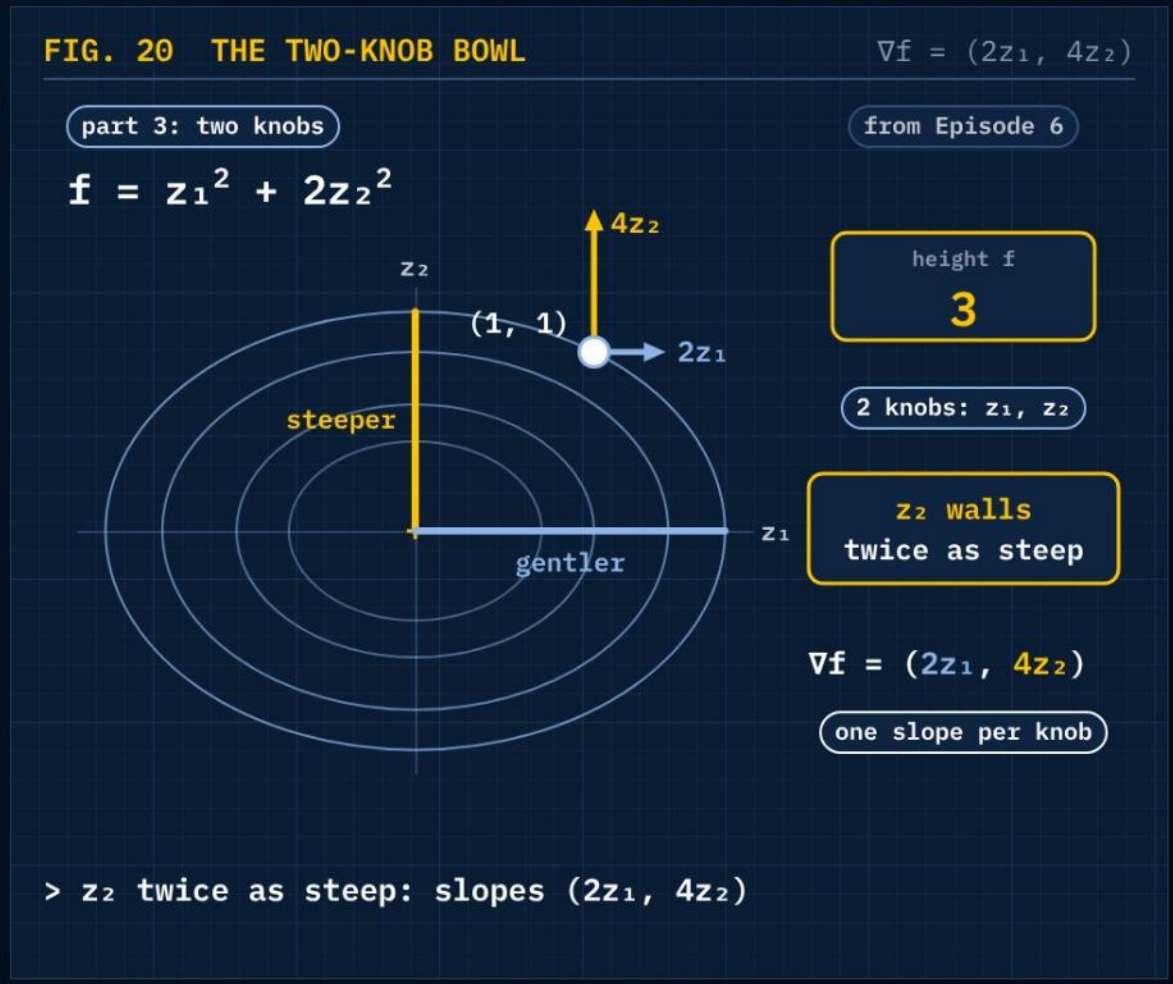
Two knobs, two steepnesses

$f = z_1^2 + 2z_2^2$, from (1, 1), height 3

01 Episode 6's bowl

02 z_2 walls are twice as steep

03 Gradient: $(2z_1, 4z_2)$



Episode 6 (the bowl). Worked example; every number is checked in the companion notebook.

PART 3 · THE BOWL

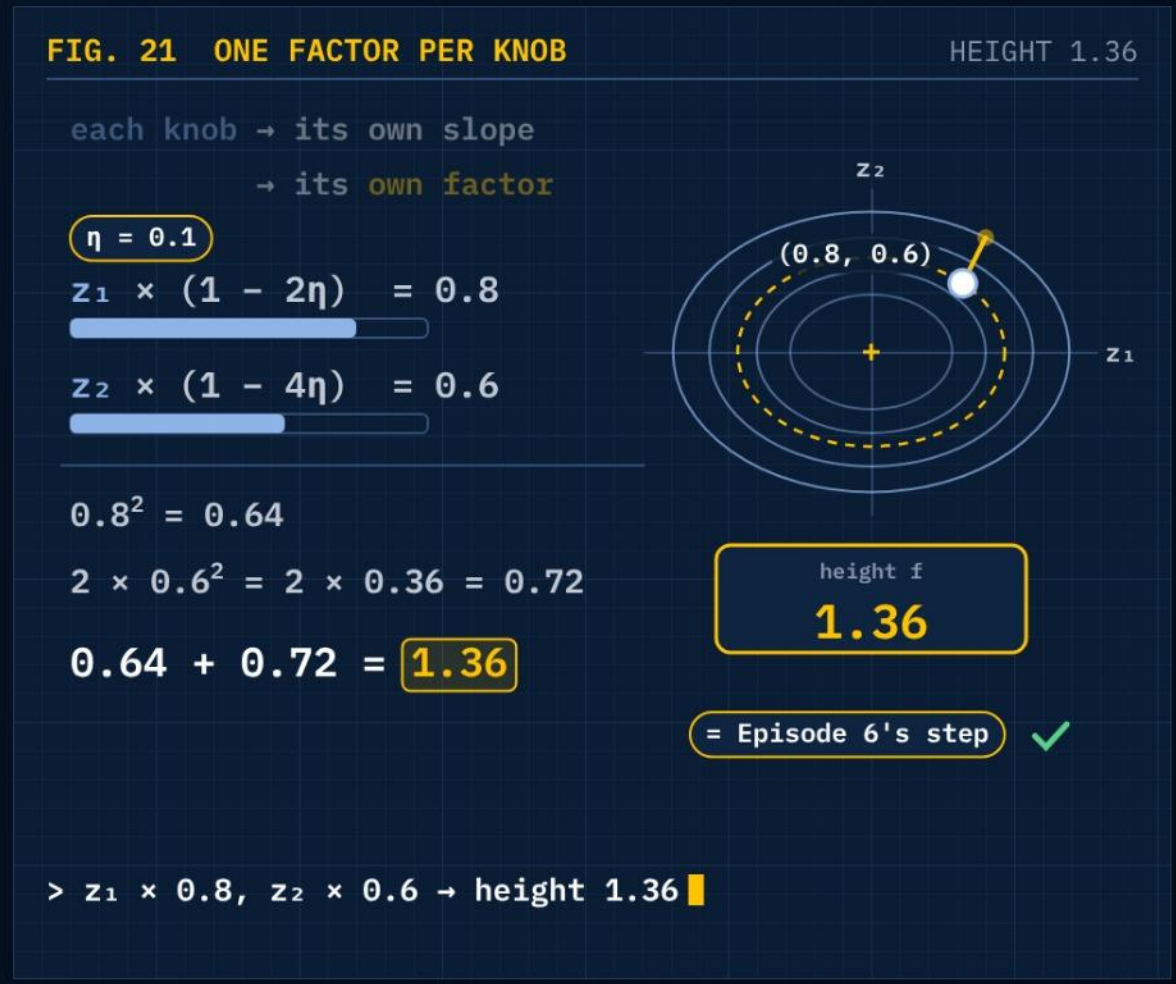
Each knob, its own factor

$z_1 \times (1 - 2\eta) \cdot z_2 \times (1 - 4\eta)$

01 Each knob only cares about its own slope

02 $\eta = 0.1$: factors 0.8 and 0.6

03 $0.64 + 2 \times 0.36 = 1.36$ (Episode 6)



Worked example; every number is checked in the companion notebook.

PART 3 · THE SPEED LIMIT

The steepest direction sets it

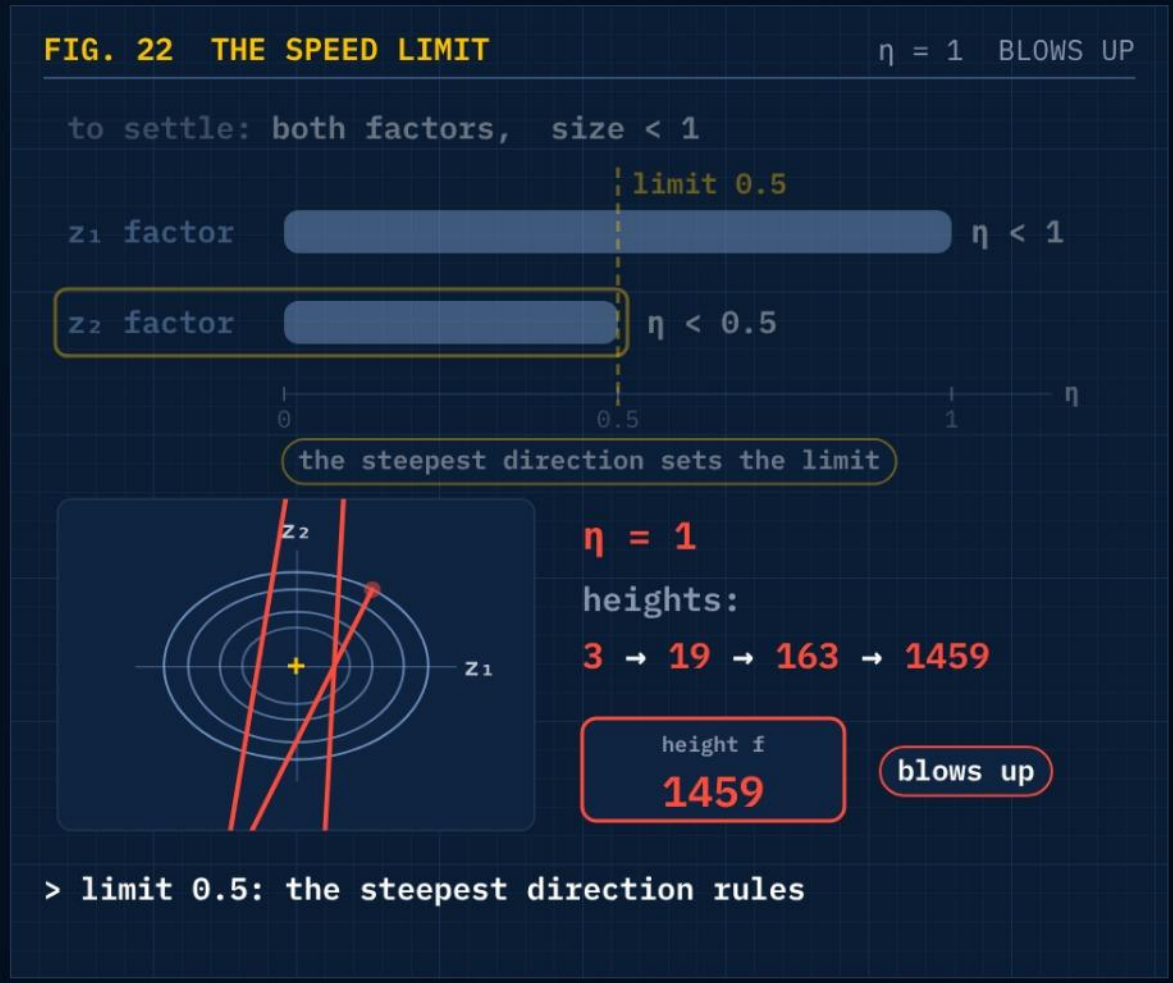
Both factors must stay smaller than 1 in size.

01 z_1 needs $\eta < 1$ · z_2 needs $\eta < 0.5$

02 Limit: 0.5, set by the steepest direction

03 $\eta = 0.5$: stuck at height 2

04 $\eta = 1$: 3, 19, 163, 1459



Worked example; every number is checked in the companion notebook.

PART 3 · ONE STEP VS MANY

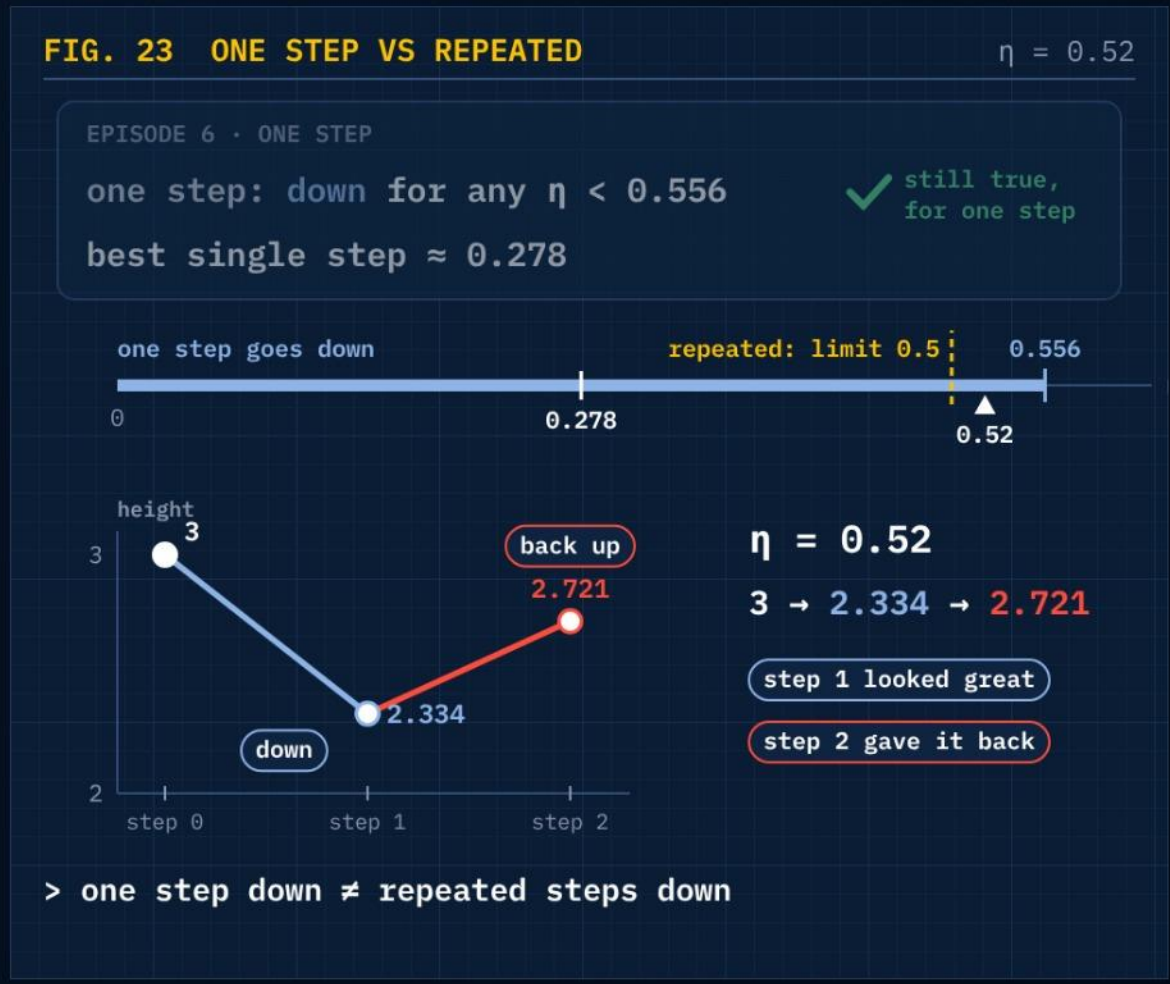
One good step isn't enough

Episode 6: one step goes down for any η < 0.556.

01 One step: best ≈ 0.278

02 Repeated: the limit is 0.5

03 η = 0.52: 3 → 2.334 → 2.721



Episode 6. Worked example; every number is checked in the companion notebook.

PART 3 · THE BEST REPEATED STEP

η = 1/3

$$1 - 2\eta = -(1 - 4\eta)$$

01 Both factors become 1/3

02 Height ÷ 9 each step: 0.333, 0.037, 0.004

03 0.278 starts faster, then falls behind

04 Five steps: both within 0.01

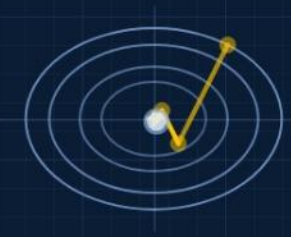
FIG. 24 BEST REPEATED STEP — 5 STEPS → 0.01

same size: $1 - 2\eta = -(1 - 4\eta)$

solve: $\eta = 1/3$

both factors: size 1/3

every step: height ÷ 9



height	step 1	step 2	step 3
η = 1/3	0.333	0.037	0.004
η = 0.278	0.222	0.039	0.008
	0.278 ahead	0.278 behind	2× as high

5 steps at 1/3: both within 0.01

> best repeated step: η = 1/3

Worked example; every number is checked in the companion notebook.

PART 3 · STRETCHED BOWLS

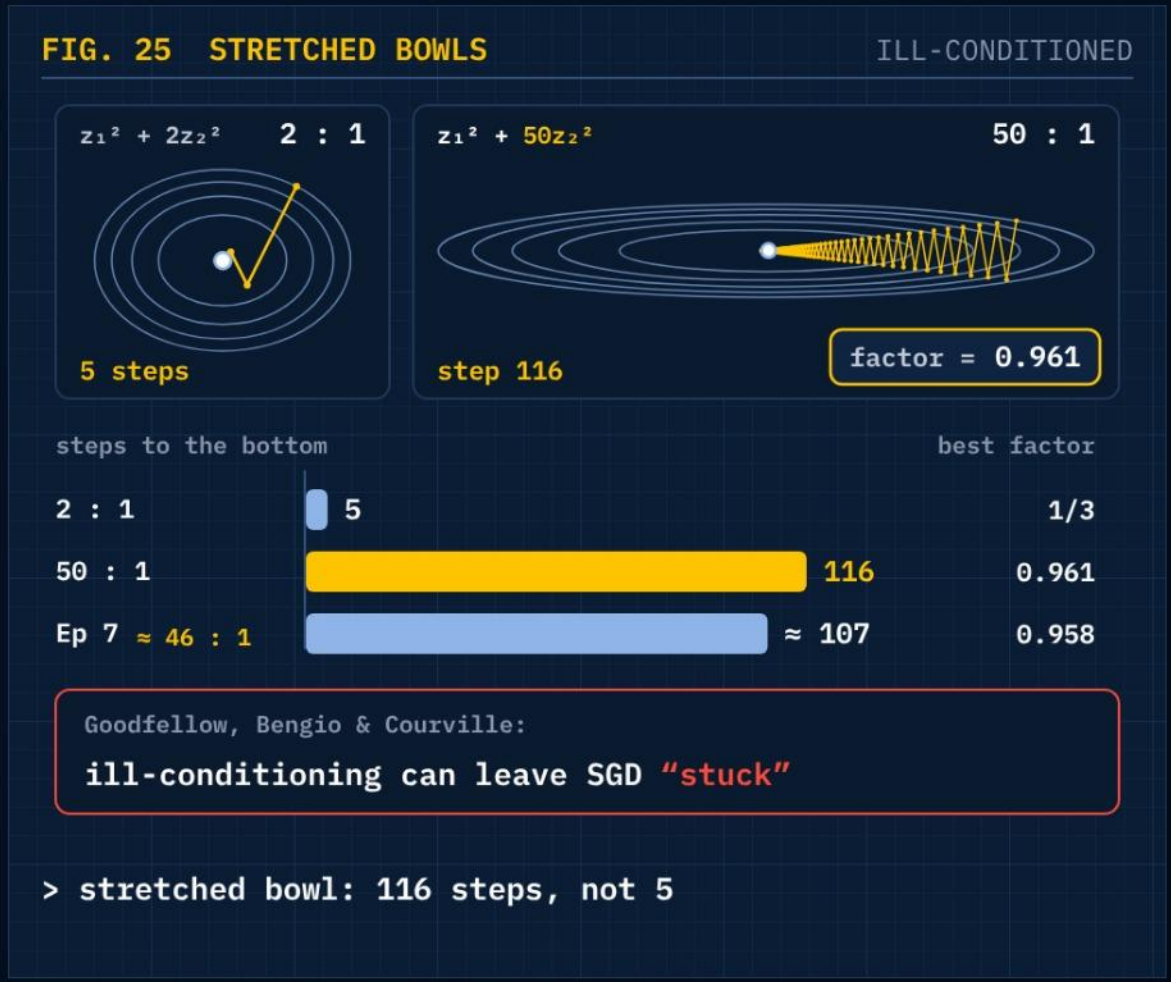
Stretched means slow

The more stretched the bowl, the slower every η.

01 50 to 1: best factor 0.961, 116 steps

02 Episode 7's loss: ~46 to 1, about 107 steps

03 Ill-conditioning can leave SGD "stuck"



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 8 §8.2.1 (ill-conditioning). Episode 7 (three-dot loss). Worked example; every number is checked in the companion notebook.

PART 3 · A PICTURE

A skateboarder in a half-pipe

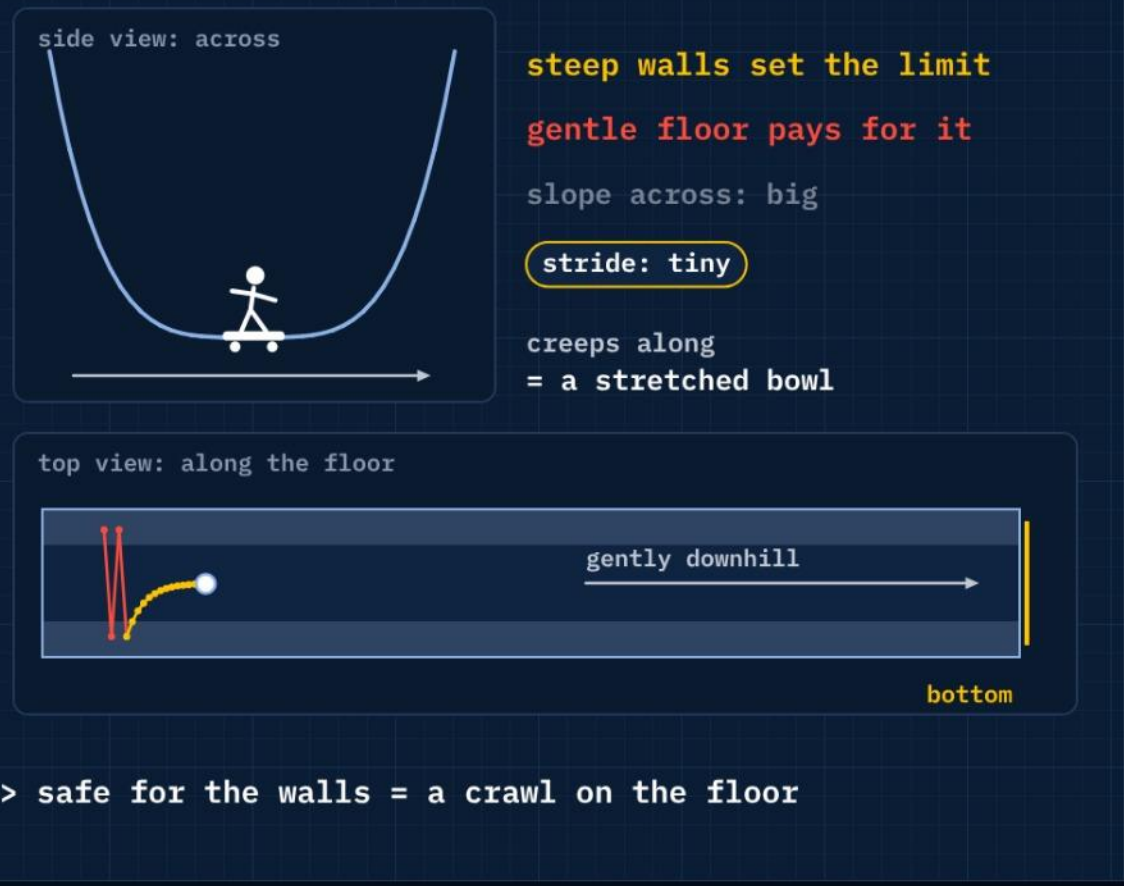
Steep walls, a gentle floor.

01 Steep walls: a big stride flies wall to wall

02 Shrink the stride: she creeps along the floor

03 The walls set the limit; the floor pays

FIG. 26 THE HALF-PIPE — STRETCHED BOWL



Analogy built for this series.

NOT EVERY LOSS IS A BOWL

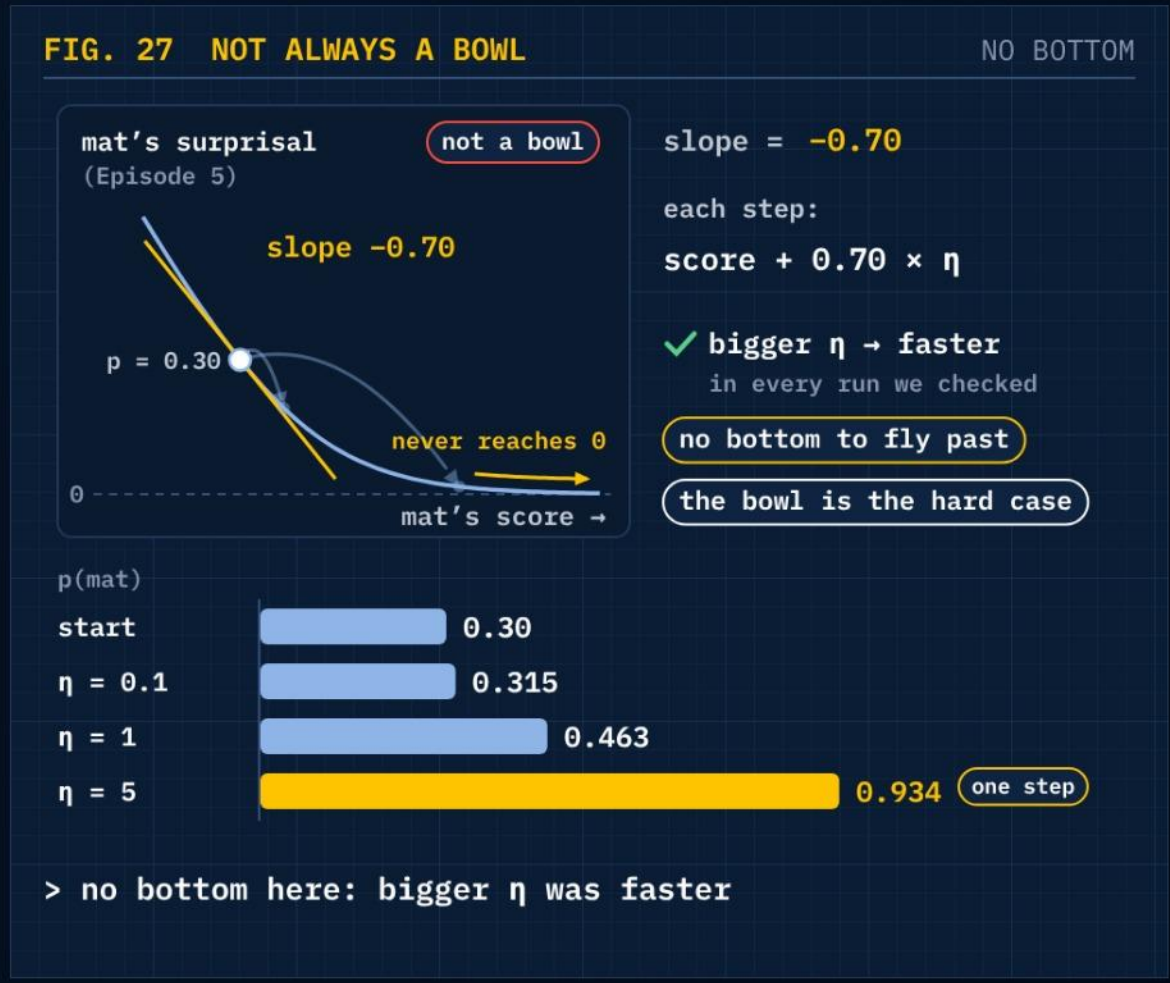
mat's surprisal

Slope −0.70: each step raises mat's score by 0.70 × η.

01 From p = 0.30 (Episode 5)

02 η = 0.1 → 0.315 · η = 1 → 0.463 · η = 5 → 0.934

03 No bottom to fly past: the bowl is the hard case



Episode 5 (mat's surprisal). The probabilities are illustrative. Worked example; every number is checked in the companion notebook.

MYTH CHECK

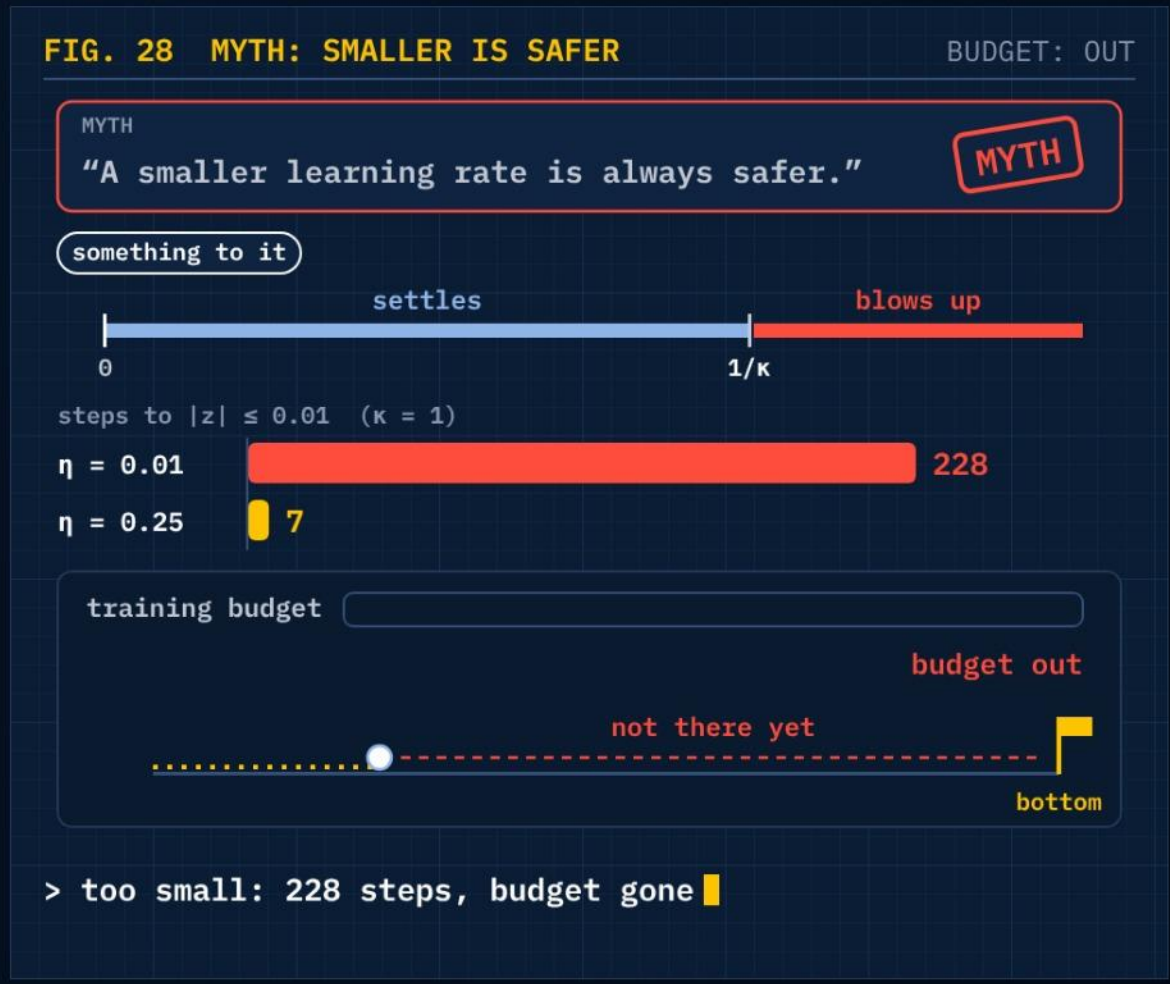
“A smaller learning rate is always safer”

Something to it: $0 < \eta < 1/\kappa$ settles.

01 Something to it: too large can blow up

02 But: 228 steps at 0.01, 7 at 0.25

03 Best bowl step was 1/3, not the smallest



Ruder, “An overview of gradient descent optimization algorithms” (arXiv:1609.04747, 2016), §§2–3. Worked example; every number is checked in the companion notebook.

RECAP

The learning rate, by hand

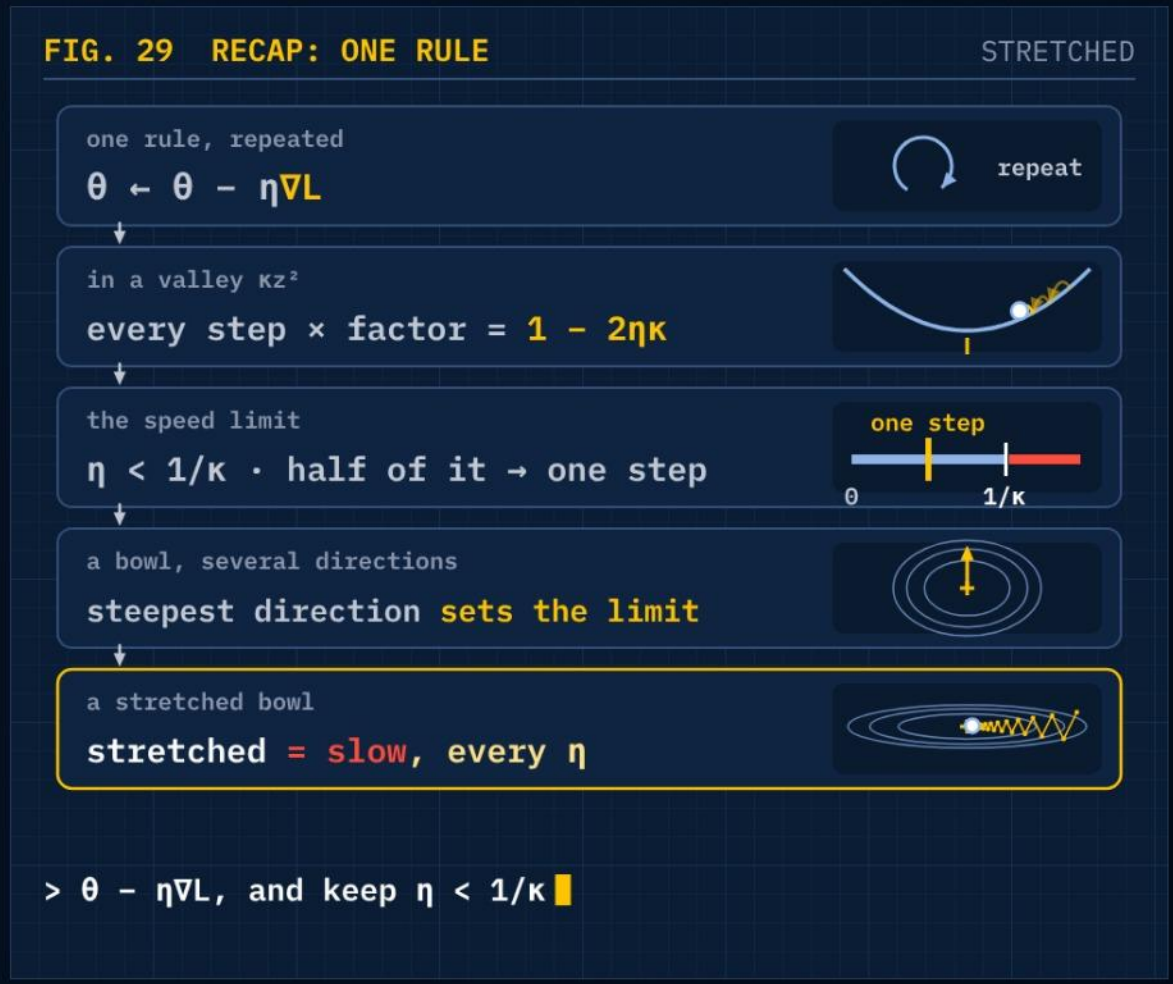
Gradient descent repeats one rule: $\theta - \eta\nabla L$.

01 Repeat $\theta - \eta\nabla L$

02 Each step multiplies by $1 - 2\eta\kappa$

03 $\eta < 1/\kappa$; half of it lands in one step

04 The steepest direction sets the limit



All sources are listed in the description.

YOUR TURN

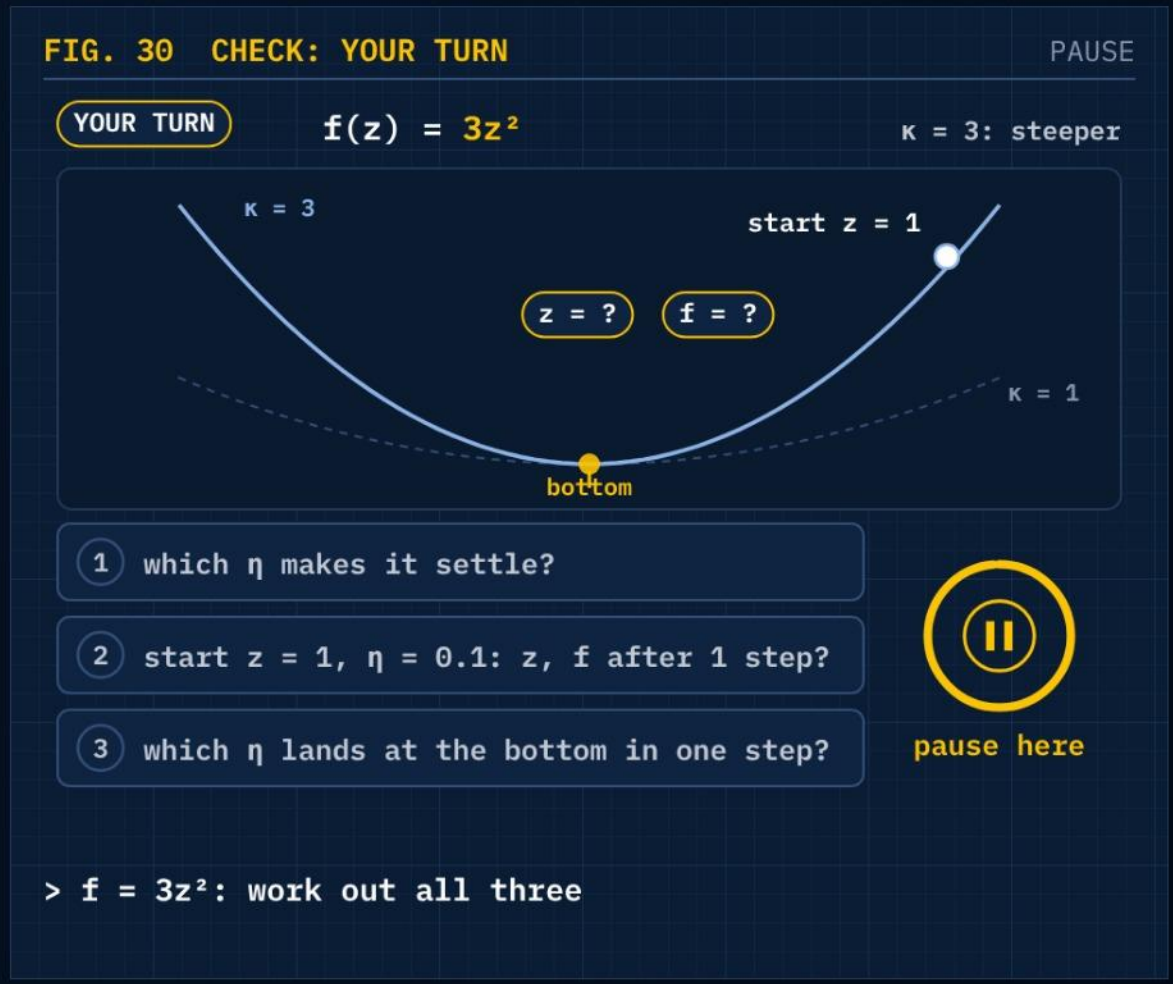
$f(z) = 3z^2$

Grab a pencil.

01 Which η settle?

02 $z_0 = 1$, $\eta = 0.1$: z and f after one step?

03 Which η lands in one step?



Worked example; every number is checked in the companion notebook.

YOUR TURN

The answer

$\kappa = 3$.

01 $0 < \eta < 1/3$

02 Factor $0.4 \rightarrow z = 0.4, f = 0.48$

03 One step: $\eta = 1/6$

FIG. 31 ANSWERS, BY HAND 3 / 3

1 $\kappa = 3 \rightarrow 0 < \eta < 1/3$ ✓

2 $\eta = 0.1$

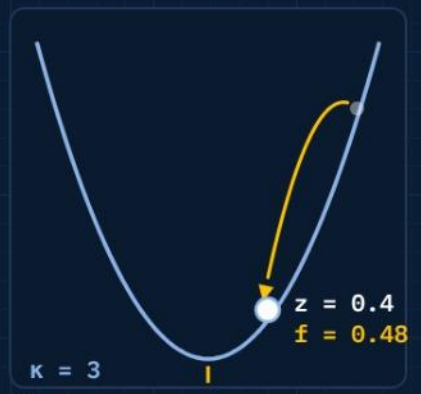
factor = 1 − 2 × 0.1 × 3

= 1 − 0.6 = 0.4

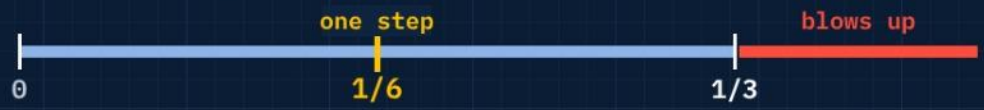
z = 0.4 ✓

f = 3 × 0.4²

= 3 × 0.16 = 0.48 ✓ down from 3



3 one step: η = ½ × 1/3 = 1/6 ✓



predict η before you press run

> κ = 3: η < 1/3, one step at η = 1/6

Worked example; every number is checked in the companion notebook.

MARK

TRY THIS WEEK

Three things to try.

01

Pick a κ, find the η that lands in one step, and check it by hand

02

In the notebook, slide η across 1/κ: settling flips to blowing up

03

On the bowl, run five steps at 0.278 and five at 1/3. Which wins?

THE MISTAKE TO AVOID

Judging a learning rate by a single step. One step can go down while the next ten go up.

Companion notebook and every source in the description.

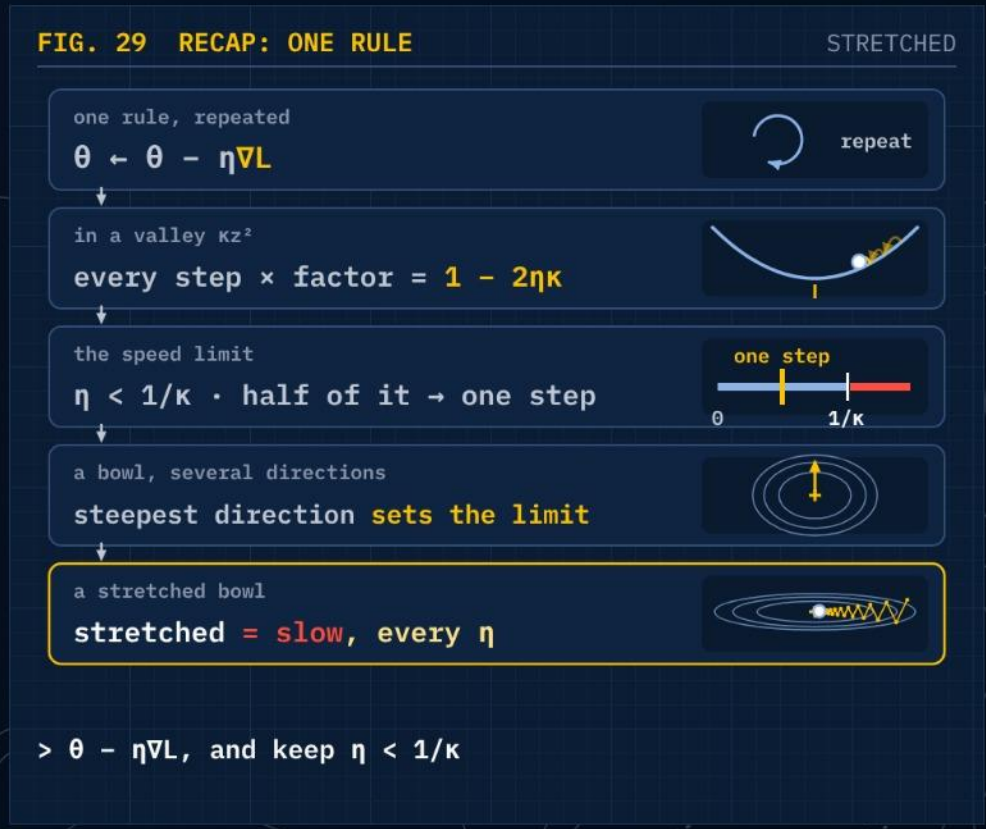
Kinvecta

NEXT TIME

Episode 10: Gradient descent, part 2. Mean vs sum, and mini-batches.

In the comments: which hiker are you? Tiny steps, giant leaps, or just right?

How Large Language Models Work · subscribe or follow for Episode 10



MARK